26900292 **MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

MG5DSCHSC300 – NUTRITION THROUGH LIFECYCLE

Duration: 2 hours **Maximum Marks: 70**

Answer Key

PART A – Short Answer Questions (2 marks each) Answer any 10 questions

1. Define EAR. **[K] / [CO1]**

Estimated Average Requirement (EAR) is the average daily nutrient intake estimated to meet the requirement of half of the healthy individuals in a particular age- and sex-specific group.

Scheme: Abbreviation – 1 mark. Definition 1 mark.

2. What is meant by dietary modification for the elderly? **[U] / [CO2]**

Dietary modification for the elderly means adapting food quantity, texture and nutrient composition to age-related changes, health conditions and chewing, swallowing or digestion difficulties.

Scheme: Meaning/adaptation of diet – 2 marks.

3. How do sex and physiological status affect RDA? **[U] / [CO1]**

Nutrient requirements vary with sex because of differences in body size and composition. Physiological states such as pregnancy and lactation increase the requirements for selected nutrients.

Scheme: Sex-related variation – 1 mark; physiological status – 1 mark.

4. What is meant by dietary modification for adults? **[U] / [CO2]**

It is the adjustment of food intake to meet an adult's energy and nutrient needs according to age, sex, activity level, health status and lifestyle, while maintaining a balanced diet.

Scheme: Definition and relevant factors – 2 marks.

5. Mention two common breastfeeding difficulties. **[U] / [CO4]**

1. Poor attachment or positioning of the baby. 2. Sore or cracked nipples (engorgement or perceived insufficient milk are also acceptable).

Scheme: Any two appropriate difficulties – 1 mark each.

6. Mention two foods that can improve the nutrient quality of the maternal diet. **[U] / [CO3]**

Examples include eggs and green leafy vegetables. Pulses, milk/curd, fruits, nuts and fish are also acceptable examples.

Scheme: Any two suitable foods – 1 mark each.

7. How does ageing affect body composition? **[U] / [CO2]**

Ageing commonly reduces muscle mass and total body water, while body fat may increase or become more centrally distributed. Bone mass may also decline.

Scheme: Any two relevant changes – 1 mark each.

8. Why should caffeine intake be limited during pregnancy? **[U] / [CO3]**

Slower Maternal Metabolism, Constriction of Blood Vessels, lBW babies, Preterm birth, pregnancy loss

Scheme: any 2- 1 mark each

9. List any two characteristics of growth during the preschool period. **[U] / [CO5]**

1. Growth is slower and steadier than during infancy. 2. Motor skills, language, social development and independence in feeding improve.

Scheme: Any two characteristics – 1 mark each.

10. Mention two principles of supplementary feeding. **[U] / [CO4]**

1. Start complementary foods at about six months while continuing breastfeeding. 2. Provide safe, age-appropriate, nutrient-dense foods with gradually increasing texture, variety and frequency.

Scheme: Any two principles – 1 mark each.

11. Why is adequate water intake important for adolescents? **[U] / [CO5]**

Water maintains hydration and temperature regulation and supports circulation, digestion and physical performance, especially during activity and hot weather.

Scheme: Any two valid functions/reasons – 1 mark each.

12. Mention two social milestones of infancy. **[K] / [CO4]**

1. Social smiling in response to a familiar face. 2. Developing attachment to caregivers and responding to familiar people (e.g., stranger anxiety later in infancy).

Scheme: Any two suitable milestones – 1 mark each.

PART B – Short Essay Questions (5 marks each) Answer any 6 questions

13. Explain how RDA serves as a guide in planning diets for healthy individuals. [U] / [CO1]

A reference for daily nutrient intake. Estimate nutrient needs while planning balanced meals. Selection and quantity of foods to meet nutrient requirements. • It helps assess whether a planned diet is likely to be adequate for most healthy individuals. • It is used with dietary guidelines, food habits, activity, affordability and individual variation; it is not a diagnostic tool for an individual.

Scheme: 1 mark for each of the five key points.

14. Explain the major food and nutritional requirements of adults. [An] / [CO2]

Energy needs should match age, sex, body size and physical activity to maintain healthy body weight. Include cereals/millets or other whole grains, pulses, vegetables, fruits, and suitable protein foods. Provide adequate high-quality protein for maintenance and tissue repair. Ensure adequate vitamins and minerals, especially calcium, iron and other nutrients according to individual needs. Include healthy fats, dietary fibre and sufficient water; limit excess salt, added sugar and saturated/trans fats.

Scheme: 1 mark for each of the five key points.

15. Explain how Reference Man and Reference Woman are used to understand nutritional requirements. [U] / [CO2]

Reference Man and Reference Woman are standardised descriptions of healthy adults used as reference models. Defined with age, body weight, health and activity level in the relevant national guidance. Estimate energy and nutrient requirements for population groups. Calculation and presentation of reference values such as recommended nutrient intakes. Guides, not exact requirements for every person; individual needs vary with body size, activity and physiological state.

Scheme: 1 mark for each of the five key points.

16. Explain the importance of nutrition during adolescent growth. [K] / [CO5]

Adolescence involves rapid increases in height, weight and tissue development, increasing energy needs. Protein supports growth and repair of body tissues. Calcium and vitamin D support bone growth and peak bone mass. Iron needs rise, particularly after menstruation begins; iron-rich foods help prevent deficiency. A varied balanced diet, supports learning, immunity and physical activity; limit nutrient-poor foods.

Scheme: 1 mark for each of the five key points.

17. Describe the role of prolactin in lactation. [U] / [CO3]

Prolactin is a hormone secreted by the anterior pituitary gland. During pregnancy, prolactin levels rise and prepare the mammary glands for milk production. After delivery, the fall in oestrogen and progesterone permits prolactin to support milk synthesis. Suckling stimulates prolactin release, helping maintain milk production. Frequent and effective breastfeeding supports continued production; prolactin produces milk.

Scheme: 1 mark for each of the five key points.

18. Explain the dietary management of pregnancy-induced hypertension. [K] / [CO3]

Balanced diet with adequate energy and protein appropriate to maternal needs. Include vegetables, fruits, whole grains and suitable protein sources; avoid highly processed foods. Avoid excessive salt. Maintain appropriate fluid. Regular antenatal monitoring.

Scheme: 1 mark for each of the five key points.

19. Describe the different anthropometric measurements used for infants. [U] / [CO4]

Weight: • Length: • Head circumference: • Weight-for-length and age-specific growth charts: Chest Circumference, Mid-upper arm circumference.

Scheme: 1 mark for each of the five key points.

20. Explain the importance of school lunch in meeting nutritional requirements. [U] / [CO5]

Share of the child's daily energy and nutrient needs. A balanced meal can provide cereals, pulses/protein foods, vegetables and other locally available nutritious foods. Helps prevent hunger and supports attention, learning and participation in school. Improve dietary diversity. Promotes healthy food habits and equity; hygiene, safe water and appropriate portion sizes are essential.

Scheme: 1 mark for each of the five key points.

PART C – Essay Questions (10 marks each), Answer any 2 questions

21. Explain how dietary guidelines and ICMR-RDA can be integrated in practical balanced-diet planning. [U] / [CO1]

Assess the individual's age, sex, activity level, physiological status, health and food preferences. Use ICMR-NIN dietary guidelines to select a varied pattern of foods. Use the relevant RDA values as reference targets. (5 marks)

Distribute foods across meals. Adjust portions and food choices. Limit excess salt, added sugar and unhealthy fats; ensure adequate fibre and water. Review the plan and modify it; (5 marks)

22. Describe the role and functions of the placenta during pregnancy. [U] / [CO3]

The placenta is a temporary organ that develops in the uterus and connects the foetus to the mother through the umbilical cord. (2 marks).

Transfers oxygen and nutrients from maternal to foetal circulation. Transfers carbon dioxide and metabolic waste products from the foetus for removal by the maternal organs. Acts as an endocrine organ, producing hormones such as hCG, progesterone, oestrogens and human placental lactogen to support pregnancy. Provides selective protection and transfers maternal IgG antibodies, contributing to passive immunity; it is not a complete barrier to pathogens or harmful substances. Anchors the developing pregnancy through its attachment to the uterine wall and supports exchange between maternal and foetal blood without normally mixing them directly.

Scheme: Definition : 2 marks, Role and function – 2 marks each (Any 4)

23. Explain the role of food habits, meal pattern and food preferences in planning a preschool diet. [U] / [CO5]

• Assess the child's usual food habits, appetite, cultural practices, family diet and any food restrictions.

• Offer regular meals and planned nutritious snacks;

Provide small, age-appropriate portions. Offer a variety of foods from major food groups. Respect preferences. Use suitable textures, colours and child-friendly. Maintain food hygiene and safety. Adjust the plan to growth, activity, appetite and nutrient needs; monitor growth over time.

Scheme: Role of food habits: 5 marks, Meal Pattern and food preferences – 5 marks

24. Discuss the protective and immunological factors present in human milk. [U] / [CO4]

• Secretory IgA provides mucosal protection by helping prevent pathogens from attaching to and entering the infant's tissues.

Lactoferrin binds iron and can inhibit the growth of some harmful bacteria.

• Lysozyme damages bacterial cell walls and contributes to antimicrobial protection.

• Human milk oligosaccharides act as prebiotics and can prevent some pathogens from attaching to the intestinal lining.

• Leukocytes and other immune components, including cytokines and protective proteins, contribute to immune defence.

• These factors support development of the infant gut microbiota and reduce the risk or severity of some infections, especially gastrointestinal infections.

• Colostrum is particularly concentrated in immune factors; human milk provides ongoing protection but does not eliminate all infection risk.

Scheme: Any 5 factors – 2 marks each